

Matrix Algebra

A Mini-Course for Multiple Regression

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Roadmap

1. Why We Need Matrix Algebra
2. Vectors and Matrices
3. Special Matrices
4. Matrix Operations
5. Vector Norms
6. The Matrix Inverse
7. Solving Linear Systems
8. OLS in Matrix Form
9. Summary

1. Why We Need Matrix Algebra

1.1 Where Lecture 7 Left Off

Lectures 6 and 7 worked with one predictor:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, \dots, n.$$

Two coefficients, n scalar equations, two separate OLS formulas.

Real data has many predictors. A more realistic model has p inputs:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i.$$

Now there are $p + 1$ coefficients and n equations.

Writing separate scalar formulas for $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ and their standard errors becomes unmanageable. We need a more compact language.

1.2 The Plan

Matrix algebra is the compact language. By the end of this lecture, the entire data set will be written in *one* matrix equation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

And in Lecture 9 the general OLS estimator will fit on one line:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

This lecture's job is to define every symbol in those two equations and the rules for moving them around. No statistics yet — just notation and arithmetic.

2. Vectors and Matrices

2.1 Scalars

A **scalar** is just a number. Single-letter symbols, lowercase: x , y , β , σ .
Lives in $\mathbb{R}$.

Examples:

- The age of a single survey respondent: $x = 27$.
- The OLS slope from Lecture 7: $\hat{\beta}_1 = 0.0475$.

Up to this point in the course, almost every quantity has been a scalar.
Now we step up to ordered collections of scalars.

2.2 Vectors

Definition — vector

A **vector** of dimension m is an ordered list of m scalars, written as a column:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} \in \mathbb{R}^m.$$

Bold symbol $\mathbf{x}$; entries $x_1, \dots, x_m$ are scalars.

Notation reminder. $\mathbb{R}^m$ is the set of all m -tuples of real numbers — the space the vector lives in.

By default, “vector” means *column* vector. A **row vector** $(x_1, x_2, \dots, x_n)$ exists too — it’s a vector written horizontally with n entries.

2.3 Vector Examples

Example 1. The (age, income, years of schooling) of one person, $m = 3$:

$$\mathbf{x} = \begin{pmatrix} 27 \\ 48000 \\ 16 \end{pmatrix} \in \mathbb{R}^3.$$

Example 2. The (height, weight, resting heart rate, blood pressure) of one patient, $m = 4$:

$$\mathbf{x} = \begin{pmatrix} 170 \\ 68 \\ 72 \\ 120 \end{pmatrix} \in \mathbb{R}^4.$$

In data work, each row of a spreadsheet (one observation) is a vector.

2.4 Matrices

Definition — matrix

An $m \times n$ **matrix** is a rectangular array of mn scalars arranged in m rows and n columns:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}.$$

Notation. The entry in row i , column j is written a_{ij} . Indices range over $1 \leq i \leq m$ and $1 \leq j \leq n$. Bold symbol $\mathbf{A}$ for the whole matrix; non-bold a_{ij} for individual entries.

Read “ $m \times n$ ” as “ m by n ”: m rows by n columns, always in that order.

2.5 Matrix Examples

Example 1. A 2×3 matrix:

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & -2 \\ 0 & 3 & 5 \end{pmatrix}.$$

Entries: $a_{11} = 1$, $a_{12} = 4$, $a_{13} = -2$, $a_{21} = 0$, $a_{22} = 3$, $a_{23} = 5$.

Example 2. A 3×2 matrix:

$$\mathbf{B} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 4 & 3 \end{pmatrix}.$$

Note: $\mathbf{A}$ and $\mathbf{B}$ have the same number of entries (6 each), but they are *not* the same matrix — their shapes differ.

2.6 A Vector Is a Matrix

Key Result — vectors are special matrices

- A **column vector** with m entries is an $m \times 1$ matrix.
- A **row vector** with n entries is a $1 \times n$ matrix.

Every rule we state for matrices applies to vectors as the special case where one dimension equals 1.

This is a pedagogical simplification: instead of separate vocabulary for vectors and matrices, we have *one* object (the matrix) with two dimensions, and vectors are the “one-dimensional-slice” case.

3. Special Matrices

3.1 Square Matrices

A matrix is **square** if it has the same number of rows and columns:
 $m \times m$.

Why care? Some operations — most importantly the matrix inverse (§6) — are defined only for square matrices.

Example of a 3×3 square matrix:

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & -1 \\ 0 & 4 & 5 \end{pmatrix}.$$

3.2 The Identity Matrix

Definition — identity matrix

The **identity matrix** $\mathbf{I}_m$ is the $m \times m$ square matrix with 1's on the diagonal and 0's everywhere else:

$$(\mathbf{I}_m)_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j. \end{cases}$$

$$\mathbf{I}_2 = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & \mathbf{1} \end{pmatrix} \quad \mathbf{I}_3 = \begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 \\ 0 & 0 & \mathbf{1} \end{pmatrix}$$

$\mathbf{I}_m$ plays the role of the number 1 for matrix multiplication: $\mathbf{A}\mathbf{I}_n = \mathbf{A}$ and $\mathbf{I}_m\mathbf{A} = \mathbf{A}$ for any $m \times n$ matrix $\mathbf{A}$ (verified in §4.5).

3.3 The Zero Matrix

The **zero matrix** $\mathbf{0}$ has every entry equal to 0. The dimensions are usually clear from context.

$$\mathbf{0}_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Plays the role of the number 0 for addition: $\mathbf{A} + \mathbf{0} = \mathbf{A}$ for any matrix $\mathbf{A}$ of matching size.

4. Matrix Operations

4.1 Equality

Two matrices **A** and **B** are **equal** when

- they have the same dimensions, and
- $a_{ij} = b_{ij}$ for every i, j .

In particular, a 2×3 matrix is never equal to a 3×2 matrix, even if they have the same numbers in some order.

4.2 Addition

Definition — matrix addition

If **A** and **B** are both $m \times n$, their sum is the $m \times n$ matrix with entries

$$(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}.$$

Addition is entry-wise. Matrices of different shapes cannot be added.

Example 1 ($2 \times 2 + 2 \times 2$).

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 0 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}.$$

Example 2 ($2 \times 3 + 2 \times 3$).

$$\begin{pmatrix} 1 & 4 & -2 \\ 0 & 3 & 5 \end{pmatrix} + \begin{pmatrix} 2 & -1 & 6 \\ 3 & 0 & -5 \end{pmatrix} = \begin{pmatrix} 3 & 3 & 4 \\ 3 & 3 & 0 \end{pmatrix}.$$

4.3 Scalar Multiplication

Definition — scalar multiplication

Given a scalar $c \in \mathbb{R}$ and an $m \times n$ matrix $\mathbf{A}$, the product $c\mathbf{A}$ is the $m \times n$ matrix with every entry of $\mathbf{A}$ multiplied by c :

$$(c\mathbf{A})_{ij} = c \cdot a_{ij}.$$

Example 1: $c = 2$, $\mathbf{A}$ a 2×2 .

$$2 \cdot \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

Example 2: $c = -1$, $\mathbf{A}$ a 2×3 .

$$-1 \cdot \begin{pmatrix} 1 & 4 & -2 \\ 0 & 3 & 5 \end{pmatrix} = \begin{pmatrix} -1 & -4 & 2 \\ 0 & -3 & -5 \end{pmatrix}.$$

4.4 Transpose

Definition — transpose

The **transpose** $\mathbf{A}^\top$ of an $m \times n$ matrix $\mathbf{A}$ is the $n \times m$ matrix obtained by swapping rows and columns:

$$(\mathbf{A}^\top)_{ij} = a_{ji}.$$

Notation reminder. Transpose is a function, $\top : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{n \times m}$. If $\mathbf{A}$ has shape $m \times n$, then $\mathbf{A}^\top$ has shape $n \times m$ — the dimensions flip.

Three properties

- $(\mathbf{A}^\top)^\top = \mathbf{A}$.
- $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$.
- $(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$ (note the order flip).

4.4 Transpose — Examples

Example 1 (transpose of a 2×3 matrix).

$$\mathbf{A} = \begin{pmatrix} 1 & 4 & -2 \\ 0 & 3 & 5 \end{pmatrix} \Rightarrow \mathbf{A}^{\top} = \begin{pmatrix} 1 & 0 \\ 4 & 3 \\ -2 & 5 \end{pmatrix}.$$

Shape goes from 2×3 to 3×2 .

Example 2 (column vector to row vector).

$$\mathbf{x} = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix} \Rightarrow \mathbf{x}^{\top} = \begin{pmatrix} 3 & 4 & 5 \end{pmatrix}.$$

Shape goes from 3×1 to 1×3 . We will use this trick a lot in §5 (norms) and §8 (OLS).

4.5 Matrix Multiplication — The Big One

Dimension rule

For $\mathbf{AB}$ to be defined: the number of columns of $\mathbf{A}$ must match the number of rows of $\mathbf{B}$.

$$\underbrace{\mathbf{A}}_{r \times s} \cdot \underbrace{\mathbf{B}}_{s \times t} = \underbrace{\mathbf{AB}}_{r \times t}.$$

Read the dimensions as a sentence: $(r \times s) \cdot (s \times t)$. The two s 's in the middle must match. The r on the left and the t on the right give the shape of the product.

If the inner dimensions don't match, the product is simply undefined.

4.5 Matrix Multiplication — Entry Formula

Entry formula

With **A** of shape $r \times s$ and **B** of shape $s \times t$, the entry in row i , column j of **AB** is

$$(\mathbf{AB})_{ij} = \sum_{k=1}^s a_{ik} b_{kj}.$$

Read this as “row i of **A** dotted with column j of **B**.”

- Take the i -th row of **A**: $(a_{i1}, a_{i2}, \dots, a_{is})$.
- Take the j -th column of **B**: $(b_{1j}, b_{2j}, \dots, b_{sj})^\top$.
- Multiply entry-by-entry and add: $a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{is}b_{sj}$.

Each entry of **AB** requires s multiplications and $s - 1$ additions. The full product has $r \cdot t$ such entries.

4.5 Matrix Multiplication — Picture

A $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ \end{bmatrix}$
row 2 of **A**

B $\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$
col 2 of **B**

AB $\begin{bmatrix} \cdot & \cdot \\ \cdot & ? \\ \end{bmatrix}$

$$(\mathbf{AB})_{22} = a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32}$$

“row 2 of **A** dot column 2 of **B**”

4.5 Matrix Multiplication — Example 1

Two 2×2 matrices.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix}.$$

Dimensions: $(2 \times 2)(2 \times 2) = 2 \times 2$. Inner 2's match. Result $\mathbf{AB}$ is 2×2 with four entries to compute.

Entry $(1,1)$. Row 1 of $\mathbf{A}$ is $(1, 2)$. Column 1 of $\mathbf{B}$ is $(5, 7)^\top$.

$$(\mathbf{AB})_{11} = 1 \cdot 5 + 2 \cdot 7 = 5 + 14 = 19.$$

Entry $(1,2)$. Row 1 of $\mathbf{A}$ is $(1, 2)$. Column 2 of $\mathbf{B}$ is $(6, 8)^\top$.

$$(\mathbf{AB})_{12} = 1 \cdot 6 + 2 \cdot 8 = 6 + 16 = 22.$$

4.5 Matrix Multiplication — Example 1 2

Entry (2,1). Row 2 of **A** is (3,4). Column 1 of **B** is (5,7)[⊤].

$$(\mathbf{AB})_{21} = 3 \cdot 5 + 4 \cdot 7 = 15 + 28 = 43.$$

Entry (2,2). Row 2 of **A** is (3,4). Column 2 of **B** is (6,8)[⊤].

$$(\mathbf{AB})_{22} = 3 \cdot 6 + 4 \cdot 8 = 18 + 32 = 50.$$

Result.

$$\mathbf{AB} = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}.$$

4.5 Matrix Multiplication — Example 2

A 2×3 times **a** 3×2 .

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 4 & 1 \\ 0 & -2 \\ 1 & 3 \end{pmatrix}.$$

Dimensions: $(2 \times 3)(3 \times 2) = 2 \times 2$. Inner 3's match. Result **AB** is 2×2 .

Entry (1,1).

$$(\mathbf{AB})_{11} = 1 \cdot 4 + 0 \cdot 0 + 2 \cdot 1 = 4 + 0 + 2 = 6.$$

Entry (1,2).

$$(\mathbf{AB})_{12} = 1 \cdot 1 + 0 \cdot (-2) + 2 \cdot 3 = 1 + 0 + 6 = 7.$$

4.5 Matrix Multiplication — Example 2 2

Entry (2, 1).

$$(\mathbf{AB})_{21} = (-1) \cdot 4 + 3 \cdot 0 + 1 \cdot 1 = -4 + 0 + 1 = -3.$$

Entry (2, 2).

$$(\mathbf{AB})_{22} = (-1) \cdot 1 + 3 \cdot (-2) + 1 \cdot 3 = -1 - 6 + 3 = -4.$$

Result.

$$\mathbf{AB} = \begin{pmatrix} 6 & 7 \\ -3 & -4 \end{pmatrix}.$$

Note: $\mathbf{BA}$ would be $(3 \times 2)(2 \times 3) = 3 \times 3$, a completely different shape.

$\mathbf{AB} \neq \mathbf{BA}$ in general — they may not even have the same shape.

4.5 Matrix Multiplication — Properties

What does work

- **Associative:** $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ (when defined).
- **Distributive:** $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$.
- **Identity:** $\mathbf{AI}_n = \mathbf{A}$ and $\mathbf{I}_m\mathbf{A} = \mathbf{A}$.

What does not work

- **Not commutative:** in general $\mathbf{AB} \neq \mathbf{BA}$. In fact $\mathbf{BA}$ may not even be defined (Example 2 above).
- **No cancellation:** $\mathbf{AB} = \mathbf{AC}$ does *not* imply $\mathbf{B} = \mathbf{C}$.

Order matters. Always write multiplications in the order they appear — never silently flip.

4.5 Four Ways to View the Same Matrix

(i) whole matrix

$$\begin{bmatrix} \mathbf{A} \end{bmatrix}$$

(ii) entry grid

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

(iii) row stack

$$\begin{bmatrix} \text{row 1} \\ \text{row 2} \\ \text{row 3} \end{bmatrix}$$

(iv) column stack

$$\begin{bmatrix} \text{col 1} \\ \text{col 2} \\ \text{col 3} \end{bmatrix}$$

Same matrix, four readings. In §8 we read $\mathbf{X}$ as n rows, one per observation. In L9 we read it as $p + 1$ columns: one intercept column and p predictor columns.

5. Vector Norms

5.1 The ℓ_2 Norm (Euclidean Length)

Definition — ℓ_2 norm

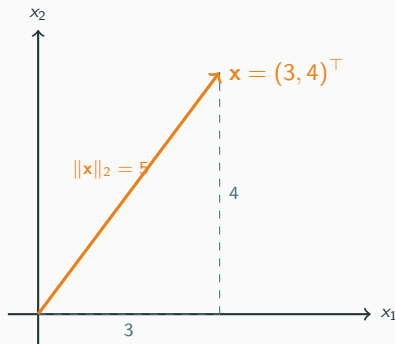
For $\mathbf{x} = (x_1, \dots, x_m)^\top$,

$$\|\mathbf{x}\|_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_m^2} = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

Reading. This is the *geometric length* of the arrow from the origin to $\mathbf{x}$. In $\mathbb{R}^2$ it is exactly the Pythagorean theorem; in $\mathbb{R}^3$ it is Pythagoras in 3D; in $\mathbb{R}^m$ it is the natural generalization.

The $\mathbf{x}^\top \mathbf{x}$ form. Recall $\mathbf{x}^\top$ is a $1 \times m$ row vector and $\mathbf{x}$ is an $m \times 1$ column vector. Their product is a 1×1 matrix — a scalar — equal to $\sum x_i^2$.

5.1 ℓ_2 Norm — Picture and Examples



Example 1: $\mathbf{x} = (3, 4)^T$. $\|\mathbf{x}\|_2 = \sqrt{9 + 16} = \sqrt{25} = 5$.

Example 2: $\mathbf{x} = (1, 2, 2)^T$. $\|\mathbf{x}\|_2 = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$.

5.2 The ℓ_1 Norm (Manhattan Length)

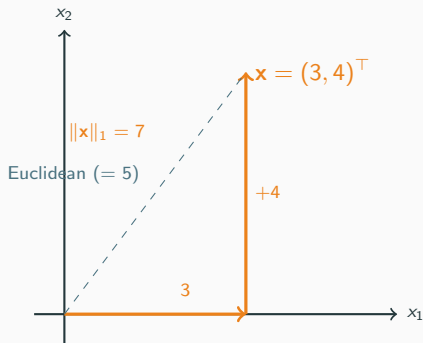
Definition — ℓ_1 norm

For $\mathbf{x} = (x_1, \dots, x_m)^\top$,

$$\|\mathbf{x}\|_1 = |x_1| + |x_2| + \dots + |x_m|.$$

Reading. Sum of absolute values. In $\mathbb{R}^2$, this is the distance a taxi drives on a city grid: it cannot cut diagonally, so it travels $|x_1|$ blocks east–west and $|x_2|$ blocks north–south.

5.2 ℓ_1 Norm — Picture and Examples



Example 1: $\mathbf{x} = (3, 4)^T$. $\|\mathbf{x}\|_1 = 3 + 4 = 7$.

Example 2: $\mathbf{x} = (-2, 1, 3)^T$. $\|\mathbf{x}\|_1 = |-2| + |1| + |3| = 2 + 1 + 3 = 6$.

5.3 Why Two Norms

Both norms measure “size” of a vector, but they show up in different places:

- ℓ_2 . The default geometric length. It measures residuals ($\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$ in §8 and L9), prediction error, and gives the ridge penalty $\sum_{j=1}^p \beta_j^2$ (L10).
- ℓ_1 . It gives the LASSO penalty $\sum_{j=1}^p |\beta_j|$ (L10), which can shrink some slope coefficients all the way to zero and produce sparse models.

L10 will show that swapping the squared-slope sum for the absolute-slope sum changes the qualitative behavior of the estimator. Both methods shrink slopes $\beta_1, \dots, \beta_p$, not the intercept β_0 .

6. The Matrix Inverse

6.1 Definition

Definition — matrix inverse

Let $\mathbf{A}$ be a square $m \times m$ matrix. Its **inverse** $\mathbf{A}^{-1}$ is the $m \times m$ matrix satisfying

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_m.$$

When such a matrix exists, $\mathbf{A}$ is called **invertible** (or **nonsingular**).

Notice the symmetry: $\mathbf{A}^{-1}$ works on either side and both products give the identity.

6.2 Intuition — Inverse Is Division

For scalars, dividing by $a \neq 0$ is the same as multiplying by $1/a = a^{-1}$. The number a^{-1} satisfies $a \cdot a^{-1} = 1$.

The matrix inverse plays the same role: $\mathbf{A}^{-1}$ “undoes” multiplication by $\mathbf{A}$, and the role of the number 1 is played by the identity $\mathbf{I}_m$.

Concrete use. If $\mathbf{Ax} = \mathbf{b}$, then multiplying both sides on the left by $\mathbf{A}^{-1}$ gives

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

This is the matrix version of $ax = b \Rightarrow x = b/a$. Section 7 walks through a concrete example; Lecture 9 Appendix A shows that the OLS normal equations use the same trick.

6.3 When Does the Inverse Exist?

Two conditions:

1. $\mathbf{A}$ must be **square** — same number of rows and columns. A rectangular matrix has no inverse in the sense defined here.
2. $\mathbf{A}$ must be **nonsingular** — informally, no column is a (linear) combination of the others. A square matrix that fails this is called **singular**.

We will not check nonsingularity by hand in this course. The computer reports whether $\mathbf{A}$ is invertible and, if so, returns $\mathbf{A}^{-1}$.

Reading. In L9, the OLS formula needs $(\mathbf{X}^\top \mathbf{X})^{-1}$ to exist. This boils down to the predictors not being redundant — a real-world issue called **multicollinearity**. Lecture 10 carries this issue forward: ridge stabilizes estimation when predictors are nearly redundant.

6.4 How Do We Compute It?

We don't. Every statistical package and every numerical library has a one-line inverse. The pedagogical work is knowing *when* to write $\mathbf{A}^{-1}$ and *what it does*, not how to find it by hand.

This mirrors the convention from Lecture 7 for the standard error: the computer plugs in the formula; we focus on the role the quantity plays.

Forward pointer. In L9 the OLS coefficient vector is

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

The expression $(\mathbf{X}^\top \mathbf{X})^{-1}$ stands for “the inverse of $\mathbf{X}^\top \mathbf{X}$, which the computer returns in one step.”

7. Solving Linear Systems

7.1 A 2×2 Linear System

Two equations in two unknowns x_1, x_2 :

$$\begin{cases} 2x_1 + 3x_2 = 8, \\ x_1 - x_2 = 1. \end{cases}$$

We will show three things in this section:

1. How to rewrite the system in matrix form $\mathbf{Ax} = \mathbf{b}$.
2. How the matrix inverse solves it.
3. Why this is the same trick used by the OLS normal equations in Lecture 9 Appendix A.

7.2 Rewrite in Matrix Form

Collect the coefficients into a matrix, the unknowns into a column vector, and the right-hand sides into a column vector:

$$\underbrace{\begin{pmatrix} 2 & 3 \\ 1 & -1 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}}_{\mathbf{x}} = \underbrace{\begin{pmatrix} 8 \\ 1 \end{pmatrix}}_{\mathbf{b}}.$$

Check. Compute the left-hand side $\mathbf{Ax}$ using the matrix-multiplication rule from §4.5:

$$\mathbf{Ax} = \begin{pmatrix} 2x_1 + 3x_2 \\ x_1 - x_2 \end{pmatrix}.$$

Setting this equal to $\mathbf{b} = (8, 1)^\top$ reproduces the two original scalar equations exactly. ✓

So the matrix form is just a compact rewriting — nothing new yet.

7.3 Solve Using the Inverse

Start from $\mathbf{Ax} = \mathbf{b}$. Multiply both sides on the left by $\mathbf{A}^{-1}$:

$$\mathbf{A}^{-1}\mathbf{Ax} = \mathbf{A}^{-1}\mathbf{b}.$$

The left side simplifies because $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_2$:

$$\mathbf{I}_2\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} \implies \mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

Key Result

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

Whenever $\mathbf{A}$ is invertible, the linear system $\mathbf{Ax} = \mathbf{b}$ has exactly one solution, given by this formula.

7.4 The Computer Reports the Inverse

For our system,

$$\mathbf{A}^{-1} = \begin{pmatrix} 0.2 & 0.6 \\ 0.2 & -0.4 \end{pmatrix}.$$

We do not derive this by hand — the computer found it.

Sanity check. Multiply $\mathbf{A}^{-1}\mathbf{A}$ using §4.5 and confirm you get $\mathbf{I}_2$:

$$\begin{pmatrix} 0.2 & 0.6 \\ 0.2 & -0.4 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0.4 + 0.6 & 0.6 - 0.6 \\ 0.4 - 0.4 & 0.6 + 0.4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \checkmark$$

7.5 Solve

Plug into $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$:

$$\mathbf{x} = \begin{pmatrix} 0.2 & 0.6 \\ 0.2 & -0.4 \end{pmatrix} \begin{pmatrix} 8 \\ 1 \end{pmatrix} = \begin{pmatrix} 0.2 \cdot 8 + 0.6 \cdot 1 \\ 0.2 \cdot 8 - 0.4 \cdot 1 \end{pmatrix} = \begin{pmatrix} 2.2 \\ 1.2 \end{pmatrix}.$$

Verify in the original system.

- $2(2.2) + 3(1.2) = 4.4 + 3.6 = 8. \checkmark$
- $(2.2) - (1.2) = 1. \checkmark$

Solution: $x_1 = 2.2$, $x_2 = 1.2$.

7.6 The Takeaway

One equation, one solution

$$\mathbf{Ax} = \mathbf{b} \implies \mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

This works for a 2×2 system, a 5×5 system, a 100×100 system — the formula is the *same*. Only the size of the matrix the computer has to invert changes.

Why we cared. The next section stacks the regression model into matrix form. Lecture 9 Appendix A then turns the OLS normal equations into $\mathbf{Ax} = \mathbf{b}$ and uses this trick to derive $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$.

8. OLS in Matrix Form

8.1 Recall the SLR Model

From Lectures 6 and 7, simple linear regression with one predictor gives n scalar equations:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n.$$

Writing out all n explicitly:

$$y_1 = \beta_0 + \beta_1 x_1 + \epsilon_1,$$

$$y_2 = \beta_0 + \beta_1 x_2 + \epsilon_2,$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 x_n + \epsilon_n.$$

n equations. Cumbersome. Time to compactify.

8.2 Stack Into Vectors and a Matrix

Collect the responses into a column vector $\mathbf{y}$, stack the predictors (with a leading 1 for the intercept) into the design matrix $\mathbf{X}$, and collect the coefficients and errors into vectors:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix}.$$

The leading column of 1's in $\mathbf{X}$ pairs with β_0 (the intercept) in the multiplication, just like the $\cdot 1$ in $\beta_0 \cdot 1 + \beta_1 x_i$.

8.3 The Compact Form

Key Result — SLR in matrix notation

$$\mathbf{y} = \mathbf{X}\beta + \epsilon$$

One equation replaces n scalar equations.

Dimensions on each side:

- $\mathbf{y}$: $n \times 1$.
- $\mathbf{X}\beta$: $(n \times 2)(2 \times 1) = n \times 1$. ✓
- ϵ : $n \times 1$.

Everything is $n \times 1$, so the matrix equation makes sense.

8.4 Verify by Row

Apply the matrix-multiplication rule from §4.5 to compute row i of $\mathbf{X}\beta$. Row i of $\mathbf{X}$ is $(1, x_i)$. The single column of β is $(\beta_0, \beta_1)^\top$:

$$(\mathbf{X}\beta)_i = 1 \cdot \beta_0 + x_i \cdot \beta_1 = \beta_0 + \beta_1 x_i.$$

Adding ϵ_i from ϵ gives $\beta_0 + \beta_1 x_i + \epsilon_i$, which equals y_i from $\mathbf{y}$. Exactly the scalar SLR equation. ✓

So the matrix form $\mathbf{y} = \mathbf{X}\beta + \epsilon$ is not a new model — it is the same n equations from §8.1, written compactly.

8.5 The $n \times (p + 1)$ Design-Matrix Convention

Convention for data problems

With n observations and p predictors:

- $\mathbf{X}$ is $n \times (p + 1)$ — one row per observation, p columns for the predictors plus one leading column of 1's for the intercept.
- $\mathbf{y}$ is $n \times 1$.
- $\boldsymbol{\beta}$ is $(p + 1) \times 1$: $(\beta_0, \beta_1, \dots, \beta_p)^\top$.
- $\boldsymbol{\epsilon}$ is $n \times 1$.

For simple linear regression ($p = 1$), $\mathbf{X}$ is $n \times 2$ — exactly what we built in §8.2.

This convention follows ISLR and will be used throughout L9, L10, and beyond.

9. Summary

9.1 Take Away — Vocabulary

The objects

- **Scalar**: a single number, $x \in \mathbb{R}$.
- **Vector**: an ordered list of m scalars, written as a column, $\mathbf{x} \in \mathbb{R}^m$.
- **Matrix**: an $m \times n$ array of scalars, $\mathbf{A} \in \mathbb{R}^{m \times n}$. Entry in row i , column j is a_{ij} .
- A vector is the special case of a matrix with one column (or one row).
- **Identity** $\mathbf{I}_m$: 1's on the diagonal, 0's elsewhere. Plays the role of 1.
- **Transpose** $\mathbf{A}^\top$: swap rows and columns; $m \times n$ becomes $n \times m$.

9.2 Take Away — Operations

The rules

- **Addition**: entry-wise; needs matching shapes.
- **Scalar multiplication**: scale every entry.
- **Matrix multiplication**: $(\mathbf{AB})_{ij}$ = row i of $\mathbf{A}$ dotted with column j of $\mathbf{B}$. Inner dimensions must match. *Not* commutative.
- ℓ_2 norm: $\|\mathbf{x}\|_2 = \sqrt{\sum x_i^2}$ = Euclidean length.
- ℓ_1 norm: $\|\mathbf{x}\|_1 = \sum |x_i|$ = Manhattan distance.
- **Inverse**: $\mathbf{A}^{-1}$ satisfies $\mathbf{AA}^{-1} = \mathbf{I}_m$. Exists for square nonsingular $\mathbf{A}$.
Computed by software.

9.3 Where Each Idea Reappears

Forward pointers

- **Lecture 9 (multiple regression)**. The general OLS estimator $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ uses every tool from this lecture: transpose, matrix product, inverse, the ℓ_2 norm in the loss $\|\mathbf{y} - \mathbf{X}\beta\|_2^2$.
- **Lecture 10 (regularization)**. Ridge regression minimizes $\|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \sum_{j=1}^p \beta_j^2$; LASSO swaps the penalty for $\sum_{j=1}^p |\beta_j|$. Both penalize slopes, not the intercept β_0 .